# 8 Session VIII: Proof Theory

## 8.1 Introduction

In this session, we introduce proof theory: the study of the mechanical manipulation of symbols. We will move away from truth and toward provability. We will explore how we can start with a set of self-evident axioms and use inference rules to calculate a proof. We will also bridge the gap between these two worlds by discussing soundness (if we can prove it, is it true?) and completeness (if it is true, can we prove it?).

In logic, the objective is to prove theorems (i.e., tautologies). This can be done in two main ways:

- Semantic methods: by evaluating truth under all possible interpretations (e.g., truth tables). This approach becomes computationally expensive and even infeasible for sufficiently expressive logics.
- Syntactic methods: by using formal, mechanical procedures that manipulate formulas based solely on their structure. These are called deductive systems, and they allow the mechanization of reasoning.

## 8.2 Semantic Tableau Method

The semantic tableau (or truth tree) method is a deductive system consisting of a formal language (propositional logic formulas) and a set of inference (decomposition) rules.

### Tree rules

In a semantic tree, we decompose complex formulas into simpler ones. The way we do this depends on whether the truth of the formula requires all sub-components to hold or just one of them.

Linear rules (non-branching): used when a formula's truth value forces a specific, singular requirement. For example, for $(V, \alpha \wedge \beta)$ to be true, both $\alpha$ and $\beta$ must be true. We list these one after the other on the same branch.

Branching rules: used when a formula's truth value offers multiple paths or possibilities. For example, for $(V, \alpha \vee \beta)$ to be true, either $\alpha$ is true or $\beta$ is true. This creates two separate branches to explore.

Each formula is labeled with a truth value: $V$ (true) or $F$ (false).

$$(V, \neg\alpha) \Rightarrow (F, \alpha), \quad (or : \tfrac{V,\neg\alpha}{F,\alpha}), \quad (F, \neg\alpha) \Rightarrow (V, \alpha), (V, \alpha \wedge \beta) \Rightarrow (V, \alpha), (V, \beta), (F, \alpha \wedge \beta) \Rightarrow \begin{cases} (F, \alpha) \\ (F, \beta) \end{cases}$$

$$(V, \alpha \vee \beta) \Rightarrow \begin{cases} (V, \alpha) \\ (V, \beta) \end{cases}, (F, \alpha \vee \beta) \Rightarrow (F, \alpha), (F, \beta), (V, \alpha \to \beta) \Rightarrow \begin{cases} (F, \alpha) \\ (V, \beta) \end{cases}$$

$$(F, \alpha \to \beta) \Rightarrow (V, \alpha), (F, \beta), (V, \alpha \leftrightarrow \beta) \Rightarrow \begin{cases} (V, \alpha), (V, \beta) \\ (F, \alpha), (F, \beta) \end{cases}, (F, \alpha \leftrightarrow \beta) \Rightarrow \begin{cases} (V, \alpha), (F, \beta) \\ (F, \alpha), (V, \beta) \end{cases}$$

### Principles of the Method

1. Proof by contradiction: to prove that a formula $\alpha$ is valid, we assume $(F, \alpha)$ and try to derive a contradiction.
2. Tree construction: we build a finite tree, where the root is labeled with $(F, \alpha)$, and each branch is expanded using the tree rules.
3. Closure condition: a branch is closed (marked $\times$) if it contains both $(V, \alpha)$ and $(F, \alpha)$ for some formula $\alpha$.

Question: Show that the formula $(A \vee B) \wedge (\neg A \wedge \neg B)$ is unsatisfiable.

1. Proof by contradiction: to prove the formula is unsatisfiable, we start by assuming it is true $(V)$. If every resulting path in our logic tree leads to a logical impossibility, the original formula must be false under all interpretations.
2. Tree construction: we decompose the formula step-by-step using the transformation rules. (a) Root: $(V, (A \vee B) \wedge (\neg A \wedge \neg B))$ (b) Apply the linear rule for AND $(\wedge)$: from the root, we derive two mandatory conditions on the same branch: $(V, A \vee B)$, $(V, \neg A \wedge \neg B)$ (c) Apply linear rule for AND again: decomposing

$(V, \neg A \wedge \neg B)$ gives us: $(V, \neg A) \Rightarrow (F, A)$ and $(V, \neg B) \Rightarrow (F, B)$ (d) Apply branching rule for OR ($\vee$): now we decompose $(V, A \vee B)$, which creates two separate paths: left branch: $(V, A)$, right branch: $(V, B)$ 3. Closure condition: we check each branch for a contradiction (having both $(V, X)$ and $(F, X)$ for any atom $X$).

**Solution 1: Semantic Tree**

$$
\begin{array}{c}
(V, (A \vee B) \wedge (\neg A \wedge \neg B)) \\
| \\
(V, A \vee B) \\
| \\
(V, \neg A \wedge \neg B) \\
| \\
(V, \neg A) \\
| \\
(V, \neg B) \\
| \\
(F, A) \\
| \\
(F, B) \\
\begin{array}{cc} / & \backslash \\ (V, A) & (V, B) \\ | & | \\ \times & \times \end{array}
\end{array}
$$

Left branch: contains $(F, A)$ and $(V, A)$.
Right branch: contains $(F, B)$ and $(V, B)$.
Result: Closed ($\times$). Unsatisfiable.

**Solution 2: Stepwise Derivation**

Step 1: $\frac{V,(A\vee B)\wedge(\neg A\wedge\neg B)}{V,A\vee B, \quad V,\neg A\wedge\neg B}$

Step 2: $\frac{V,\neg A\wedge\neg B}{V,\neg A, \quad V,\neg B}$

Step 3: $\frac{V,\neg A}{F,A}$ and $\frac{V,\neg B}{F,B}$

Step 4 (Branching): $\frac{V,A\vee B}{(V,A) \quad \text{or} \quad (V,B)}$

Branch 1: $(V, A)$ and $(F, A) \Rightarrow \times$
Branch 2: $(V, B)$ and $(F, B) \Rightarrow \times$

All branches are closed,
so the formula is unsatisfiable.

## 8.3 Formal Proofs Using Modus Ponens

Truth tables verify tautologies by checking all possible cases. This is analogous to verifying a numerical identity by testing all values individually. For example, instead of checking: $\forall n \leq 2^{63}, \quad \sum_{k=1}^{n} k^3 = \left(\sum_{k=1}^{n} k\right)^2$

case by case, we can prove it using mathematical induction. Similarly, in propositional logic, we use formal proofs based on axioms and inference rules. We check the smallest possible value for $n$ ($n = 1$). Assume the identity holds for some arbitrary $n = m$. We must prove that if it holds for $m$, it must hold for $m + 1$. This is the "Modus Ponens" of induction $(m, m \to m + 1)$.

**Axioms** (Hilbert-style)

$$
\begin{aligned}
&A1 : \alpha \to (\beta \to \alpha) \\
&A2 : (\alpha \to (\beta \to \gamma)) \to ((\alpha \to \beta) \to (\alpha \to \gamma)) \\
&A3 : (\neg\alpha \to \neg\beta) \to ((\neg\alpha \to \beta) \to \alpha)
\end{aligned}
$$

**Inference rule (Modus Ponens)** $\frac{\alpha \quad \alpha\to\beta}{\beta}$

Any tautology can be proven using only these three formulas and the rule of Modus Ponens. None of these three can be proven from the other two; they are all necessary.
A1 (Law of Simplification): If $\alpha$ is true, then it is true regardless of whether $\beta$ is true. It allows you to introduce extra assumptions that don't change the truth of your original statement.
A2 (Distributive Law of Implication): This is the "engine" of the system. It allows you to distribute an assumption ($\alpha$) over a conditional ($\beta \to \gamma$).
A3 (Law of Clavius / Reductio ad Absurdum): This handles negation. It states that if assuming $\neg\alpha$ leads to both $\neg\beta$ and $\beta$ (a contradiction), then $\alpha$ must be true.

Other logicians have developed different "Axiom Schemas" to achieve the same result. For example: Whitehead and Russell (Principia Mathematica) used a different set of 5 axioms. Nicod's Axiom proved that we can actually reduce all of propositional logic to one single (very long) axiom if you use the "NAND" operator. Extended Logics: If we move beyond basic propositional logic into modal logic

(dealing with necessity $\Box$ and possibility $\Diamond$), we add new axioms like: Axiom K: $\Box(P \to Q) \to (\Box P \to \Box Q)$

**Formal proof**: A formal proof of a formula $\alpha$ is a finite sequence $S_1, S_2, \ldots, S_n$ such that: $S_n = \alpha$, then, For each $i$ $(1 \leq i \leq n)$: either $S_i$ is an instance of an axiom, or $S_i$ is obtained from previous formulas using Modus Ponens.

**Theorem**: In formal logic (syntax), a formula $\alpha$ is a theorem (written as $\vdash \alpha$) if and only if there exists a formal proof for it.

We term $\vdash \alpha$ : $\alpha$ is provable (a theorem); $\alpha_1, \ldots, \alpha_n \vdash \beta$ : $\beta$ is derivable from the assumptions.

$\vdash \alpha$ (empty left side): This means $\alpha$ can be proven using only the axioms. This is the strongest possible status; it means $\alpha$ is a universal truth of the system (a logical theorem).

$\beta \vdash \alpha$: read as "given the assumption $\beta$, we can prove $\alpha$." $\beta$ is a hypothesis, and $\alpha$ is the conclusion.

$\{P, P \to Q\} \vdash Q$: This means "Given the assumptions $P$ and $P \to Q$, we can prove $Q$."

In general, $\Gamma \vdash \alpha$ where $\Gamma$ is a set of assumptions.

Example: Proof of $\vdash \alpha \to \alpha$ (read as "It is provable that alpha implies alpha" or "alpha implies alpha is a theorem"). To prove this theorem *purely* in a Hilbert-style system, we use Axioms A1 and A2 along with Modus Ponens (MP).

$$
\begin{aligned}
&S_1 : [\alpha \to ((\alpha \to \alpha) \to \alpha)] \to ([\alpha \to (\alpha \to \alpha)] \to (\alpha \to \alpha)) && \text{(A2)}\\
&S_2 : \alpha \to ((\alpha \to \alpha) \to \alpha) && \text{(A1)}\\
&S_3 : [\alpha \to (\alpha \to \alpha)] \to (\alpha \to \alpha) && \text{(MP: } S_1, S_2\text{)}\\
&S_4 : \alpha \to (\alpha \to \alpha) && \text{(A1)}\\
&S_5 : \alpha \to \alpha && \text{(MP: } S_3, S_4\text{)}
\end{aligned}
$$

Question: Example: Proof of $\vdash \alpha \to (\beta \to \beta)$

$$
\begin{aligned}
&S_1 : \beta \to (\beta \to \beta) && \text{(A1)}\\
&S_2 : \beta \to ((\beta \to \beta) \to \beta) && \text{(A1)}\\
&S_3 : [\beta \to ((\beta \to \beta) \to \beta)] \to \big([\beta \to (\beta \to \beta)] \to (\beta \to \beta)\big) && \text{(A2)}\\
&S_4 : [\beta \to (\beta \to \beta)] \to (\beta \to \beta) && \text{(MP: } S_3, S_2\text{)}\\
&S_5 : \beta \to \beta && \text{(MP: } S_4, S_1\text{)}\\
&S_6 : (\beta \to \beta) \to (\alpha \to (\beta \to \beta)) && \text{(A1)}\\
&S_7 : \alpha \to (\beta \to \beta) && \text{(MP: } S_6, S_5\text{)}
\end{aligned}
$$

## 8.4 Deduction Theorem (DT)

Theorem:

$$\alpha_1, \ldots, \alpha_n \vdash \beta \ \text{ iff } \ \alpha_1, \ldots, \alpha_{n-1} \vdash \alpha_n \to \beta$$

It means that a proof with assumptions can be transformed into a proof of an implication. In advanced logic, we use the DT to avoid long writing. If we know $\alpha \vdash \beta$, we just say "by the deduction theorem, $\vdash \alpha \to \beta$" and move on, making the proof much shorter.

Corollary:

$$\alpha_1, \ldots, \alpha_n \vdash \beta \ \text{ iff } \ \vdash \alpha_1 \to (\alpha_2 \to \cdots (\alpha_n \to \beta) \cdots)$$

Example: Proof of transitivity (if $P \to Q$ and $Q \to R$, then $P \to R$). In a pure Hilbert system, this is a nightmare to write ( 15-20 steps). But with the deduction theorem, we can turn a complex implication into a simple chain of facts (5 steps).

The goal: we want to prove the transitivity of implication: $\{\alpha \to \beta, \beta \to \gamma\} \vdash \alpha \to \gamma$

Step 1: use the deduction theorem "Assumption", the DT states that if we want to prove a conditional $\alpha \to \gamma$, we can temporarily move the antecedent $\alpha$ to the left side of the turnstile as a hypothesis and prove the consequent $\gamma$ instead.

New Goal: $\{\alpha \to \beta, \beta \to \gamma, \alpha\} \vdash \gamma$

Step 2: the formal sequence, with $\alpha$ available as a fact, we can reach $\gamma$ using a simple sequence of Modus Ponens (MP):

| Step | Formula | Justification |
|---|---|---|
| $S_1$ | $\alpha$ | Hypothesis (added via DT) |
| $S_2$ | $\alpha \to \beta$ | Hypothesis |
| $S_3$ | $\beta$ | MP $(S_1, S_2)$ |
| $S_4$ | $\beta \to \gamma$ | Hypothesis |
| $S_5$ | $\gamma$ | MP $(S_3, S_4)$ |

Step 3: apply the DT "Conclusion", since we have established that $\{\alpha \to \beta, \beta \to \gamma, \alpha\} \vdash \gamma$, the DT meta-rule concludes that the original goal is satisfied: $\{\alpha \to \beta, \beta \to \gamma\} \vdash \alpha \to \gamma$

Question: Everyday life example!

Imagine you are a detective. You have three pieces of evidence, and you want to prove a final "If/Then" rule.
Evidence 1 ($\alpha \to \beta$): If the safe was opened, the alarm must have been disabled.
Evidence 2 ($\beta \to \gamma$): If the alarm was disabled, the security guard was asleep.
Evidence 3 ($\gamma \to \delta$): If the security guard was asleep, the dog did not bark.
The Goal: Prove that "If the safe was opened, the dog did not bark" ($\vdash \alpha \to \delta$).
Step 1: The assumption strategy. Instead of wrestling with axioms A1 and A2 to build a 20-line proof, we use the DT. We pretend for a moment that the safe was actually opened ($\alpha$). Then Step 2: The proof sequence using MP. We get $S_7$, $\delta$, MP $(S_5, S_6)$ - The dog did not bark.
Step 3: The discharge (applying the theorem) The DT says: "Since you proved $\delta$ by assuming $\alpha$, you have now earned the right to say $\alpha \to \delta$ without needing the assumption anymore."

## 8.5 Consistency and Completeness

A logical system is inconsistent if there exists a formula $\beta$ such that: $\vdash \beta$ and $\vdash \neg\beta$.
A set $\Gamma$ is inconsistent if: $\Gamma \vdash \beta$ and $\Gamma \vdash \neg\beta$. Therefore, if $\Gamma \vdash \beta$, then $\Gamma \cup \{\neg\beta\}$ is inconsistent.
Theorem (soundness: everything provable is true)

$$\vdash \beta \quad \Rightarrow \models \beta$$

It means that every provable formula is logically valid.
Corollaries: (a) Propositional logic is consistent. (b) If $\Gamma$ is satisfiable, then $\Gamma$ is consistent.

A logical system is complete if every valid formula (tautology) is provable.
Theorem (completeness: everything true is provable)

$$\models \beta \quad \Rightarrow \vdash \beta$$

i.e., the axioms and inference rules are sufficient to derive all logically valid formulas.

In logic, syntax (what we can write/derive) and semantics (what is true/valid) are two different worlds. They meet in the middle through two specific properties: soundness and completeness. In the world of semantics, we use the symbol $\models$, focusing on meaning and interpretation, and use truth tables and valuations as a method to answer, "Is it true in every possible world (Tautology)?" In the world of syntax, we use the symbol $\vdash$, focusing on form and rules, and use semantic trees, axioms, and resolution as methods to answer, "Can we reach the goal by just following the mechanical rules?"

## 8.6 Summary

Besides the semantic tableau and the Hilbert-style system, several other formal proof methods are widely used. Natural deduction (MP-based); sequent calculus, which uses the cut rule $\frac{\Gamma \vdash A \quad A, \Delta \vdash B}{\Gamma, \Delta \vdash B}$; and type-theoretic approaches based on the Curry–Howard correspondence $f : A \to B,\ a : A \ \Rightarrow\ f(a) : B$. The resolution method, introduced in the next session, is a specific syntactic procedure for proving propositional entailments. It can be viewed as a mechanized version of the semantic tree method. While sequent calculus is good for proof theory, resolution is good for automation/SAT/security proofs.

# 9 Session IX: Clausal Logic and the Resolution Method

## 9.1 Introduction

In the previous sessions, we explored various normal forms and proof theory for propositional logic. We now turn our attention to clausal logic, a specialized form of propositional logic that represents formulas as sets of clauses. Clausal logic provides a convenient and standardized structure that is particularly suited for algorithmic reasoning.

The resolution method is one of the most important inference techniques in automated reasoning. It allows us to determine whether a given set of clauses entails a particular goal formula, bridging the gap between syntactic manipulations of clauses and the semantic notion of entailment. By transforming formulas into conjunctive normal form (CNF) and systematically applying the resolution rule, we can construct proofs of unsatisfiability, or equivalently, prove that a goal logically follows from a set of premises.

## 9.2 Clausal Form and Resolution Rule

Resolution operates on formulas written as sets of clauses. A clause set is simply a conjunction of clauses: $C_1 \wedge C_2 \wedge \cdots \wedge C_n$
Example: $(p \vee q) \wedge (\neg p \vee r) \wedge (\neg r)$
In clause-set notation, this becomes $\{\{p, q\}, \{\neg p, r\}, \{\neg r\}\}$

The empty clause, written $\square$ or $\perp$, contains no literals and represents a contradiction. If the empty clause can be derived, the entire clause set is inconsistent.

Before applying resolution, formulas must be converted into clausal form. For example, the formula $\neg(p \to q) \vee r$ should be converted to CNF: $(p \vee r) \wedge (\neg q \vee r)$; then clause set is $\{\{p, r\}, \{\neg q, r\}\}$.

The resolution rule combines two clauses containing complementary literals.
Let $C_1 = (l \vee A)$, $C_2 = (\neg l \vee B)$ where $l$ is a literal and $A, B$ are disjunctions of other literals.

The rule produces the clause $(A \vee B)$, this new clause is called the resolvent. Formally:

$$\frac{l \vee A \quad \neg l \vee B}{A \vee B}$$

The complementary literals $l$ and $\neg l$ cancel each other, and the remaining literals are merged.
Example 1: $\{p, q\}, \{\neg p, r\}$, resolving on $p$ gives $\{q, r\}$.
Question: Everyday life example of resolution!
Imagine you are deciding what to wear based on two pieces of advice: Clause 1 ($l \vee A$): "Either it rains ($l$), or I will wear my sunglasses ($A$). "Clause 2 ($\neg l \vee B$): Either it does not rain ($\neg l$), or I will bring an umbrella ($B$)." The Resolution: Since the weather must be one or the other (it either rains or it doesn't), the "Rain" variable ($l$ and $\neg l$) is eliminated. You are left with a new, certain truth: Resolvent ($A \vee B$): "I will wear my sunglasses ($A$) or I will bring an umbrella ($B$). "No matter what the clouds do, one of those two things ($A$ or $B$) must happen.
Example 2 (Modus Ponens): $\{p\}, \{\neg p, q\}$, resolvent: $\{q\}$
Question: Everyday life example of resolution!
Resolution is also the engine behind "If-Then" logic. Let $p$ be "The pizza shop is open." Clause 1 ($\neg p \vee q$): "If the pizza shop is open ($p$), then we eat pizza ($q$)." (In logic, $p \to q$ is the same as $\neg p \vee q$). Clause 2 ($p$): "The pizza shop is open." The Resolution: The "Open" status ($p$ and $\neg p$) cancels out. Resolvent ($q$): "We eat pizza."
Example 3 (contradiction): $\{p\}, \{\neg p\}$, resolvent: $\square$ (the empty clause indicates inconsistency.)
Question: Everyday life example of resolution!
Imagine you are trying to book a meeting room, and you receive two facts from different departments: Clause 1 ($p$): "The conference room is available at 10:00 AM. "Clause 2 ($\neg p$): "The conference room is not available at 10:00 AM." The Resolution: In logic, we try to resolve these by canceling the complementary literals. But here, there are no other literals ($A$ or $B$) left to carry on the story. Resolvent ($\square$): A complete contradiction. The AI system does not just say "I do not know." It concludes that the entire set of assumptions is broken. If these were clues in a mystery, the detective would know that at least one witness is lying, because both statements cannot exist in the same reality.

## 9.3 Logical Consequence (semantic entailment)

Let $\Gamma$ be a set of premises and $G$ a goal formula. The notation $\Gamma \models G$ means that $G$ is a logical consequence of $\Gamma$: every interpretation that makes all formulas in $\Gamma$ true also makes $G$ true. In other words, if all premises in $\Gamma$ hold, then $G$ must necessarily hold as well.

Semantic entailment $\Gamma \models G$ is a meta-logical statement. It asserts that $G$ is true in every interpretation in which all formulas in $\Gamma$ are true. It is about logical consequence from a set of premises, not just the truth of a single formula.

Example 1: Let $\Gamma = \{R \to W,\ R\}$ and $G = W$; then $\Gamma \models W$ because in every interpretation where $R \to W$ and $R$ are both true, $W$ must also be true.

Example 2: $P \to Q,\ \neg Q \models \neg P$

| $P$ | $Q$ | $\neg Q$ | $P \to Q$ | $\neg P$ |
|---|---|---|---|---|
| $V$ | $V$ | $F$ | $V$ | $F$ |
| $V$ | $F$ | $V$ | $F$ | $F$ |
| $F$ | $V$ | $F$ | $V$ | $V$ |
| $F$ | $F$ | $V$ | $V$ | $V$ |

Question: Material Implication $\to$ vs Semantic Entailment $\models$

There is a formal connection: $\Gamma \models G$ iff $(\bigwedge_{F\in\Gamma} F) \to G$ is valid. or $P \models Q \iff \models (P \to Q)$

So, proving semantic entailment is equivalent to proving that the implication "all premises imply the goal" is true in all interpretations.

| | Material Implication $\to$ | Syntactic Turnstile $\vdash$ | Semantic Entailment $\models$ |
|---|---|---|---|
| Name | Implication | Turnstile | Double Turnstile |
| Domain | Syntax | Syntax | Semantics |
| Type | Single formula | Meta-logical statement | Meta-logical statement |
| Meaning | $P \to Q$ is true if $P$ is false or $Q$ is true | Provable: we can derive it using mechanical rules (A1, A2, A3, MP) | Valid: it is true in all rows of a truth table |
| Scope | One conditional statement | Reasoning from a set of premises | Reasoning from a set of premises |
| Truth info | Evaluates in one interpretation at once | Evaluates derivations syntactically, without referring to truth assignments | Considers all interpretations of $\Gamma$ at once |
| Validity | Can be true in some interpretations; it does not mean that $Q$ logically follows from $P$ | A statement $\phi$ is validly derivable from $\Gamma$ if $\Gamma \vdash \phi$ | The relationship $\Gamma \models G$ holds if and only if the material implication of all premises to the goal is a tautology |

Soundness: If $\Gamma \vdash \alpha$, then $\Gamma \vDash \alpha$. (The proof is not just a game; it tracks truth). Completeness: If $\Gamma \vDash \alpha$, then $\Gamma \vdash \alpha$. (If the truth table says it follows, a proof exists).

## 9.4 Resolution as a Refutation Method

Truth tables provide a straightforward way to evaluate logical formulas, but they become impractical when the number of variables grows. A formula with $n$ variables requires a table with $2^n$ rows. For even moderately large values of $n$, this approach becomes computationally infeasible.

Resolution provides an alternative; instead of enumerating all interpretations, we attempt to derive a contradiction using a purely symbolic rule of inference. Resolution is a syntactic tool ($\vdash$). However, the only reason we use it is to determine semantic entailment ($\models$).

The task of proving $\Gamma \models G$ is central in logic, mathematics, and computer science. For example, in automated reasoning or program verification, we often know a set of facts or rules ($\Gamma$) and want to check if a particular statement or property ($G$) logically follows from them.

An intuitive analogy is that of a detective: $\Gamma$ represents all known clues, and $G$ is a hypothesis. If assuming the opposite of the hypothesis ($\neg G$) leads to a contradiction with the clues, then the hypothesis must be true. Resolution formalizes this idea by attempting to derive a contradiction from $\Gamma \cup \{\neg G\}$; if successful, it proves that $\Gamma \models G$.

**Theorem 1**: $\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ if and only if $\alpha_1, \alpha_2, \ldots, \alpha_{n-1} \models \alpha_n \to \beta$

**Theorem 2**: Let $\Gamma$ is a set of formulas. If $\Gamma \models \alpha$ and $\Gamma \models \alpha \to \beta$ then $\Gamma \models \beta$

**Theorem 3**: If $\Gamma \models \beta$ and $\Gamma \models \neg\beta$ then $\Gamma$ is unsatisfiable.

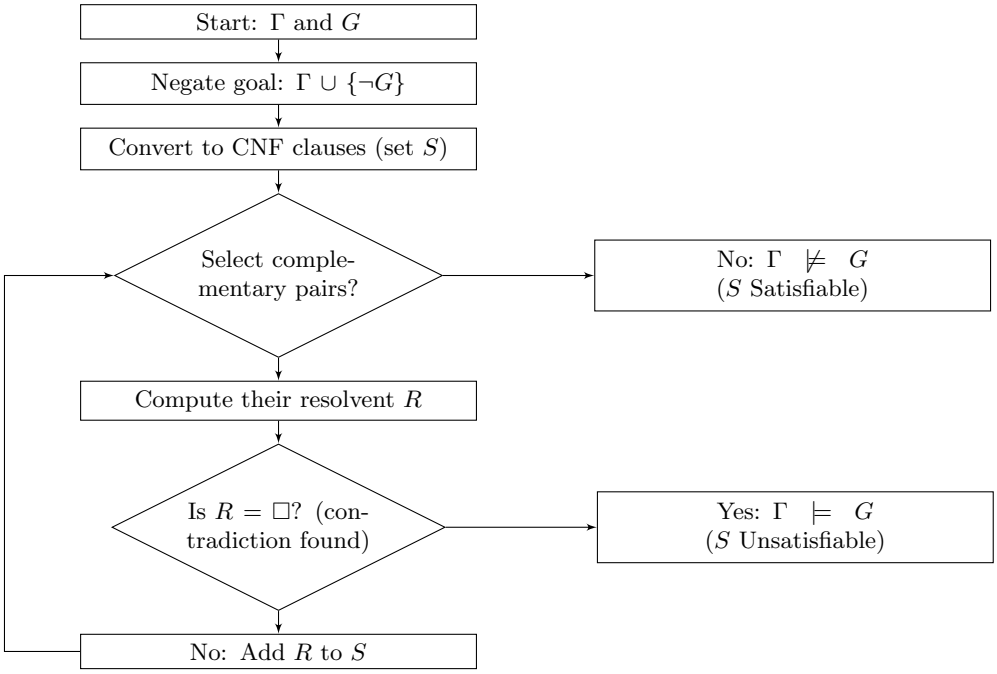
**Theorem 4**: $\alpha_1, \alpha_2, \ldots, \alpha_n \models \beta$ if and only if $\{\alpha_1, \alpha_2, \ldots, \alpha_n, \neg\beta\}$ is unsatisfiable.

Resolution is typically used as a refutation procedure: instead of proving directly that a conclusion follows from premises, we attempt to derive a contradiction.

Question: Does resolution replace the truth table?

Truth tables explicitly enumerate all models of a formula (if they exist), whereas resolution is a refutation method whose goal is to prove that no model exists. If the resolution process fails to derive a contradiction, the formula is satisfiable (it has at least one model), although the specific model is not explicitly constructed. So, in a specific scenario, if we want to know where F is true or where exactly F is false (which interpretation), resolution does not work.

To prove $\Gamma \models G$, we perform the following steps:




Example: Prove that $(p \wedge q) \to p$ is valid.

Step 1: Negate the formula $\neg((p \wedge q) \to p)$. Step 2: Simplify $(p \wedge q) \wedge \neg p$

Step 3: Clause set $\{\{p\}, \{q\}, \{\neg p\}\}$. Step 4: Resolution $\{p\}, \{\neg p\} \Rightarrow \square$

Since the empty clause is obtained, the original formula is valid.

Question: Examples of valid/invalid entailment

1. Prove that if we have $p \to q$ and $p$, then $q$ must be true.

$\Gamma$ (Premises): $\{p \to q, p\}$, $G$ (Goal): $q$. Negate goal: $\Gamma \cup \{\neg q\} = \{p \to q, p, \neg q\}$. Convert to CNF (Set $S$): * $p \to q$ becomes $(\neg p \vee q)$. $S = \{\{\neg p, q\}, \{p\}, \{\neg q\}\}$. Select complementary pairs: Pick $\{\neg p, q\}$ and $\{p\}$. Compute resolvent $R$: The $p$ and $\neg p$ cancel out. $R = \{q\}$. Is $R = \square$? No. Add $\{q\}$ to $S$. New $S = \{\{\neg p, q\}, \{p\}, \{\neg q\}, \{q\}\}$. Select next pair: Pick $\{q\}$ and $\{\neg q\}$. Compute resolvent $R$: The $q$ and $\neg q$ cancel out. Is $R = \square$? Yes. Result: $\Gamma \models G$ (The argument is valid).

2. If we have $p \vee q$ and we know $p$, can we prove $q$? (Intuitively, no; $p$ being true does not force $q$ to be true).

$\Gamma$ (Premises): $\{p \vee q, p\} G$ (Goal): $q$. Negate goal: $\Gamma \cup \{\neg q\} = \{p \vee q, p, \neg q\}$. Convert to CNF (Set $S$): $S = \{\{p, q\}, \{p\}, \{\neg q\}\}$. Select complementary pairs: Pick $\{p, q\}$ and $\{\neg q\}$. Compute resolvent $R$: The $q$ and $\neg q$ cancel out. $R = \{p\}$. Is $R = \square$? No. Add $\{p\}$ to $S$. New $S = \{\{p, q\}, \{p\}, \{\neg q\}\}$. (Note: $\{p\}$ was already there, so $S$ has not effectively changed). Select complementary pairs? There are no other different complementary pairs left to resolve that would produce new information. Result: No more pairs. The set $S$ is satisfiable (e.g., if $p = True$ and $q = False$, all clauses in $S$ are true). Conclusion: $\Gamma \not\models G$ (The argument is invalid).

3. Let the clause set be: $\Gamma = \{\, (p \vee \neg l),\ (q \vee l),\ (\neg p \vee r),\ (\neg q \vee r),\ \neg r \,\}$

resolution rule:

$\frac{p \vee \neg l \quad q \vee l}{p \vee q}$, $\frac{p \vee q \quad \neg p \vee r}{q \vee r}$, $\frac{q \vee r \quad \neg q \vee r}{r}$, $\frac{r \quad \neg r}{\square}$

## 9.5 Summary

In this session, we introduced the resolution method, a powerful rule of inference used in automated reasoning. By applying resolution repeatedly, we can detect contradictions and determine whether a logical argument is valid. This technique forms the theoretical foundation of SAT solvers, automated theorem provers, and logic programming systems. Because the number of possible clauses over a finite set of variables is finite, the resolution procedure always terminates in propositional logic (decidability).